train input, full runs — small-8way

L1D 8 sets x 8, L2 64 sets x 8 · metric schema 9 · gem5 TimingSimpleCPU · 4 November 2026

31 of 32 runs completed, 1 running, 0 pending; **1 remaining**.

Program (does not depend on the cache)

run	configs done	spill/ROI	spill ld/load	spill st/store	(spill+csave)/ROI	(spill+csave) ld/load	(spill+csave) st/store
cactuBSSN	1/1	21.33%	51.32%	55.49%	21.43%	51.50%	55.81%
parest	1/1	0.18%	0.45%	1.28%	0.49%	0.93%	7.38%
deepsjeng	1/1	2.99%	9.41%	15.98%	8.19%	23.80%	48.29%
leela	1/1	0.43%	1.55%	2.04%	6.08%	15.99%	46.72%
namd	1/1	3.22%	8.11%	17.74%	3.23%	8.13%	17.81%
lbm	1/1	9.69%	27.97%	35.99%	9.69%	27.97%	35.99%
nab	1/1	7.61%	25.91%	37.53%	8.29%	27.38%	44.95%
nab_gcn4dna	1/1	7.30%	24.89%	35.20%	8.25%	26.98%	45.65%
xz	1/1	1.36%	4.25%	4.82%	6.14%	16.52%	26.27%
xz_combined	1/1	2.79%	10.48%	25.95%	4.09%	14.43%	41.80%
imagick	1/1	0.04%	0.21%	4.19%	0.25%	0.82%	48.78%
blender	1/1	0.31%	0.55%	4.03%	3.18%	5.93%	39.04%
povray	1/1	0.27%	0.46%	0.95%	14.26%	22.21%	55.40%
omnetpp	1/1	0.01%	0.02%	0.03%	12.59%	24.64%	48.38%
xalancbmk	1/1	0.44%	1.00%	1.04%	18.29%	32.42%	63.05%
x264	1/1	6.30%	13.42%	40.78%	8.75%	18.41%	57.31%
gcc_200	1/1	1.16%	2.64%	4.34%	16.68%	35.31%	67.49%
gcc_scilab	1/1	1.25%	2.90%	4.70%	15.74%	33.65%	65.08%
gcc	1/1	1.41%	3.09%	5.21%	16.29%	32.84%	65.64%
mcf	1/1	1.61%	2.27%	13.28%	2.75%	4.04%	21.88%
perlbench_scrabbl	1/1	0.23%	0.30%	0.93%	17.65%	32.90%	56.14%
perlbench_suns	1/1	0.98%	2.06%	2.86%	18.85%	36.45%	59.85%
perlbench_diffmail	1/1	2.06%	4.51%	4.76%	17.54%	31.15%	53.97%
perlbench_perfect	1/1	1.44%	2.34%	5.00%	18.85%	33.31%	60.43%
perlbench_splitmail	1/1	5.79%	9.36%	20.68%	17.56%	30.55%	58.86%
bwaves_1	1/1	2.99%	7.65%	18.86%	2.99%	7.65%	18.86%
bwaves_2	1/1	3.04%	7.82%	18.83%	3.04%	7.82%	18.83%
exchange2	1/1	4.84%	10.67%	12.41%	6.03%	13.09%	15.72%
fotonik3d	1/1	3.99%	10.66%	12.65%	4.00%	10.69%	12.75%
roms	0/1	—	—	—	—	—	—
wrf	1/1	0.92%	3.44%	2.61%	1.59%	4.97%	8.09%
cam4	1/1	2.49%	9.15%	7.96%	4.59%	14.65%	19.40%
pooled		3.60%	9.44%	17.30%	7.34%	17.15%	41.91%
mean		3.18%	8.35%	13.49%	9.59%	20.87%	41.47%

Miss rates

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
cactuBSSN	done	15.63%	5.02%	35.06%	4.67%	13.03%	64.55%	12.21%	7.58%
parest	done	10.03%	1.20%	10.73%	1.24%	1.75%	71.14%	5.24%	16.99%
deepsjeng	done	8.41%	1.80%	10.80%	7.31%	4.48%	7.44%	2.65%	13.44%

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
leela	done	4.79%	0.25%	5.52%	0.20%	2.81%	21.89%	0.03%	85.88%
namd	done	11.48%	2.41%	14.60%	1.25%	2.44%	28.27%	9.44%	17.93%
lbm	done	21.37%	0.00%	7.38%	0.46%	74.88%	97.30%	—	32.18%
nab	done	4.04%	0.11%	6.39%	0.18%	0.56%	7.34%	0.55%	1.90%
nab_gcn4dna	done	4.04%	0.08%	6.31%	0.27%	0.49%	71.26%	20.08%	3.32%
xz	done	5.01%	0.08%	7.72%	1.39%	0.80%	45.61%	0.00%	6.56%
xz_combined	done	9.70%	1.02%	12.52%	0.74%	4.12%	23.25%	0.05%	2.37%
imagick	done	5.01%	34.95%	5.00%	0.40%	1.76%	94.98%	1.01%	5.81%
blender	done	6.67%	4.89%	7.28%	4.40%	2.74%	12.85%	1.81%	10.36%
povray	done	8.58%	6.97%	10.50%	19.27%	3.66%	48.92%	67.90%	72.66%
omnetpp	done	15.90%	0.05%	21.21%	0.73%	5.49%	29.84%	97.51%	3.06%
xalancbmk	done	11.40%	12.13%	14.65%	23.99%	4.81%	13.19%	0.00%	0.04%
x264	done	1.85%	0.15%	2.26%	0.85%	1.84%	33.89%	3.40%	15.49%
gcc_200	done	7.83%	0.90%	10.45%	4.05%	3.21%	43.28%	5.65%	9.85%
gcc_scilab	done	8.43%	1.15%	11.28%	3.78%	3.39%	43.14%	3.62%	10.75%
gcc	done	8.17%	0.98%	10.89%	4.01%	3.21%	46.10%	5.28%	10.13%
mcf	done	22.97%	4.49%	26.71%	7.12%	7.31%	77.82%	6.96%	19.34%
perlbench_scrabbl	done	5.68%	4.63%	8.10%	9.31%	1.52%	14.81%	0.00%	0.01%
perlbench_suns	done	4.98%	0.17%	7.12%	5.35%	1.57%	22.02%	0.60%	20.40%
perlbench_diffmail	done	4.49%	0.13%	6.21%	5.79%	1.61%	42.24%	0.27%	10.70%
perlbench_perfect	done	7.71%	0.63%	10.89%	5.22%	2.31%	57.56%	41.87%	27.88%
perlbench_splitmail	done	4.02%	0.04%	5.90%	2.07%	1.50%	21.83%	1.89%	3.31%
bwaves_1	done	24.89%	4.88%	25.55%	0.09%	39.60%	36.09%	3.57%	0.50%
bwaves_2	done	24.95%	4.75%	25.60%	0.09%	40.03%	36.24%	3.63%	0.49%
exchange2	done	1.08%	1.02%	1.45%	2.04%	0.45%	2.98%	0.36%	1.01%
fotonik3d	done	8.73%	2.49%	11.99%	0.01%	0.13%	79.16%	2.89%	29.04%
roms	running	—	—	—	—	—	—	—	—
wrf	done	6.62%	6.01%	6.56%	4.45%	6.97%	56.91%	9.74%	13.09%
cam4	done	9.86%	1.52%	10.23%	1.90%	11.53%	61.38%	13.45%	35.85%
pooled		8.99%	2.81%	10.81%	3.21%	6.11%	45.97%	10.00%	9.47%
mean		9.49%	3.38%	11.51%	3.96%	8.06%	42.36%	10.72%	15.74%

Formulas

column	formula	aciklama
spill/ROI	`((spill_load + spill_store) / roi_total_insts) * 100`	ROI komutlarinin yuzde kaci spill load veya spill store
spill ld/load	`(spill_load / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload
spill st/store	`(spill_store / roi_store) * 100`	ROI store'larinin yuzde kaci spill store
(spill+csave)/ROI	`((spill_load + spill_store + csave_load + csave_store) / roi_total_insts) * 100`	ayni oran, callee-save save/restore dahil
(spill+csave) ld/load	`((spill_load + csave_load) / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload veya callee-save restore
(spill+csave) st/store	`((spill_store + csave_store) / roi_store) * 100`	ROI store'larinin yuzde kaci spill store veya callee-save save
L1D	`(L1D misses / L1D accesses) * 100`	Tum load ve store access'lerinin (spill + other) L1D miss rate'i: L1D'nin bu konfigurasyondaki genel performansi.
spill ld L1	`(spill load L1D misses / spill load L1D accesses) * 100`	Spill load'larin yuzde kaci L1D'de miss oldu. Genelde dusuktur: spill slot'u kisa sure once yazildigi icin cache line cogu zaman hala L1D'dedir.
other ld L1	`((load misses - spill load misses) / (load accesses - spill load accesses)) * 100`	Other load'larin (spill olmayan load'lar) L1D miss rate'i. spill_ld_L1miss ile yan yana bakilir: spill load'lar normal veriden daha mi az, daha mi cok miss oluyor?

column	formula	aciklama
spill st L1	`(spill store L1D misses / spill store L1D accesses) * 100`	Spill store'larin yuzde kaci L1D'de miss oldu. Cache write-allocate: miss olan bir store once cache line'i L2'den getirir, sonra yazar.
other st L1	`((store misses - spill store misses) / (store accesses - spill store accesses)) * 100`	Other store'larin (spill olmayan store'lar) L1D miss rate'i; spill_st_L1miss ile karsilastirmak icin.
L2	`(L2 misses / L2 accesses) * 100`	L2'ye gelen tum isteklerin (L1D ve L1I miss'leri) yuzde kaci L2'de de miss oldu ve DRAM'e gitti.
spill ld L2	`(spill load L2 misses / spill load L2 accesses) * 100`	L1D'de miss olan spill load'larin yuzde kaci L2'de de miss oldu. Payda sadece L1D'yi miss eden spill load'lardir, butun spill load'lar degil.
spill st L2	`(spill store L2 misses / spill store L2 accesses) * 100`	L1D'de miss olan spill store'larin yuzde kaci L2'de de miss oldu (payda yine sadece L1D miss'leri).

spill = compiler-confirmed register-allocator spill (GCC `-dP`), csave = callee-saved register save/restore (CFI), both inside the ROI only. pooled = sum of numerators / sum of denominators over the completed runs; mean = average of the per-run percentages. Miss rates are per cache access; an L2 rate's denominator is the L1D misses of that access type.